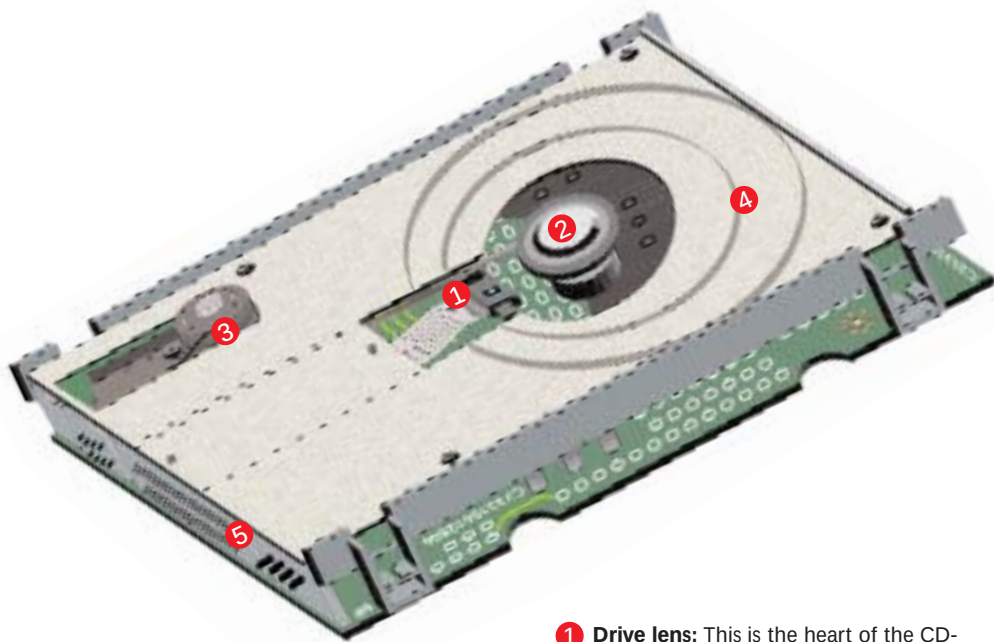




- 1 **Drive lens:** This is the heart of the CD-RW drive. It not only reads the inserted disc, but also writes to a blank disk.
- 2 **Disk motor:** This controls the motion of the disk. As the laser pickup assembly moves towards the outer edge of the disc, the motor must slow down, so that the laser can scan the disk at a constant speed.
- 3 **Tracking motor:** The tracking motor takes care of the laser assembly, and makes it read from the inner to the outer tracks on the CD.
- 4 **Drive tray:** This is the tray where you place the CD. Some drives have a slot interface, where the tray is absent.
- 5 **Connectors:** The IDE and power connectors are on the back of the drive, where you connect the cables.

Facts

- You may not be able to write at high speeds using older systems, since they are incapable of feeding data to the CD writer at the required rate. This also occurs due to slower hard drives, low processing power and limited RAM.
- If you use an ATAPI (IDE) CD writer, make sure it's on a different IDE channel from the hard drive. PCs come with at least two IDE channels—primary and secondary—built into the motherboard. Each channel supports two IDE devices—a master and a slave—on a single cable. Hard drives should always be on the primary channel, and ATAPI devices on the secondary channel. It's preferable to connect the CD writer as the secondary master device.
- Use Track-At-Once instead of Disk-At-Once to burn an audio CD. This makes the audio CD sound more professional, since a 2 second gap will be inserted between each burned track. The CD can then be produced error-free.
- Use branded CDs to burn critical data. Cheap, ordinary media are prone to damage, and cannot hold data for long.



Buying Tips

- ✓ **SPEED:** Make sure that the drive has high spin speed and low access time. Access time is more important if you need to read non-sequential data, such as in games.
- ✓ **FEATURES:** It should support oversizing, overburning, TAO, DAO, multi-session and incremental packet writing, and RAW writing.
- ✓ **BUNDLED SOFTWARE:** It should include feature-rich CD-burning software, along with drivers for all major operating systems. The audio and IDE cables should also be included.
- ✓ **FIRMWARE:** The drive's firmware should be upgradeable for future support.

Quick Find

YOU NEED	A device that lets you take backups and burn different types of CDs
Basic Use	
LOOK FOR	A decent 52X CD writer
YOU NEED	A multi-purpose drive that not only burns CDs, but can play DVDs as well
Inter Use	
LOOK FOR	A combination drive that gives good performance

The Future

Theoretically, CD writers have reached a saturation point, from where further advances seem impossible. Several vendors have invented new ways to crunch more data into a regular 700 MB disk. However, CDs burnt in this manner have been plagued with incompatibility issues with existing CD-ROM drives, which are incapable of reading such 'oversized' media.

On the other hand, DVD writers are still in a nascent stage. Currently, available DVD-Rs can store about 4 GB of data. The latest entry into this field—Blue Ray disks—can store a mammoth 23 GB of data. The future seems destined to see several such small-size, high storage media. Expect to see a slew of writers that cater to disks of this kind.